

GLOSSARY

Unit 1 Robotics

- actuator** (*n*) – a mechanical device that moves or controls parts of the robot.
- AI module** (*phr*) – a system that enables robots to process data and make intelligent decisions.
- alloy** (*n*) – a mixture of metals.
- antennae** (*n*) – a component used for communication and signal reception.
- anxiety** (*n*) – a feeling of worry, nervousness.
- appealing** (*adj*) – attractive, interesting, or pleasing.
- automata** (*n*) – mechanical devices made to imitate human actions.
- bargain** (*n*) – a deal or agreement that is good value for money.
- blueprint** (*n*) – a detailed plan used to build something.
- bolt** (*n*) – a metal fastener.
- bolt from the blue** (*idiom*) – an unexpected event or innovation.
- boatload** (*n*) – a large quantity of something (informal).
- bomb disposal** (*phr*) – the process of detecting, disabling, or safely removing explosive devices.
- boulder** (*n*) – a large, heavy rock.
- chassis** (*n*) – the main supporting base of a robot, housing essential components.
- circuit** (*n*) – wires that connect parts and carry signals or power.
- cobot** (*n*) – a collaborative robot designed to safely work alongside humans.
- cog in the machine** (*idiom*) – a small but essential part of a larger system.
- companionship** (*n*) – the feeling of friendship or closeness with another person.
- compressed** (*adj*) – pressed tight, like air in a tank.
- cooling system** (*phr*) – a mechanism that prevents overheating in robotic components.
- cybercriminal** (*n*) – a person who commits crimes using computers or the internet.
- defense** (*n*) – protection against attack or danger.
- degrees of freedom** (*phr*) – the number of independent movements a robot can make.
- despair** (*n*) – the complete loss of hope.
- domestic robot** (*phr*) – a robot designed to assist with household tasks.
- effector** (*n*) – a part of the robot (e.g., gripper, claw, etc.) that interacts with the environment.
- elderly care** (*phr*) – assistance and support provided to older adults.
- empathy** (*n*) – the ability to understand and share another person's feelings or emotions.
- extraterrestrial** (*adj*) – relating to things that come from outside Earth.
- fetch** (*v*) – to go and bring something back.
- forge** (*v*) – to shape metal.
- frame** (*n*) – the body or skeleton that holds the robot together.
- gear** (*n*) – a toothed wheel that helps parts move together.
- glitch** (*n*) – a small fault or malfunction in a system, usually temporary.
- grease the wheels** (*idiom*) – to make a process smoother or more efficient, often with tech adjustments.
- grripper** (*n*) – a hand-like part that grabs objects.
- guardian** (*n*) – someone that protects.
- housing** (*n*) – the outer cover that protects the robot's insides.
- hydraulic** (*adj*) – operated by the pressure of liquid.
- immortal** (*adj*) – never dying or lasting forever.
- inevitable** (*adj*) – certain to happen and impossible to avoid.
- ingenious** (*adj*) – very clever and original, showing creativity.
- intruder** (*n*) – a person or thing that enters a place without permission.
- joint** (*n*) – a rotating connection that allows movement in elbows or knees.

locomotion (*n*) – the ability of a robot to move from one place to another.

malfunction (*v*) – to stop working correctly.

microcontroller (*n*) – a small computer on a chip that controls specific functions in a robot.

mimic (*v*) – to copy or imitate someone or something.

motor (*n*) – a part that powers movement, like spinning wheels.

navigation system (*phr*) – a system that helps the robot move and avoid obstacles (e.g., GPS).

nursing home (*n*) – a residential facility that provides medical and personal care to elderly people who need assistance with daily living.

off the rails (*idiom*) – to go out of control or deviate from the plan.

payloads (*n*) – the items or equipment carried and delivered by a robot, vehicle, or drone.

piston (*n*) – a a movable part inside a cylinder that slides to move robot parts.

plentiful (*adj*) – existing in large amounts.

pneumatic (*adj*) – powered by compressed air or gas.

power supply (*phr*) – the energy source for the robot (e.g., battery, wired connection).

precursor (*n*) – an early stage or ingredient that leads to something later.

processor (CPU) (*phr*) – the “brain” that thinks and controls the robot.

propulsion (*n*) – the force or system that drives movement, especially in vehicles or robots.

proximity sensor (*phr*) – a sensor that detects how close an object is without physical contact.

pump (*n*) – a device that moves liquids or gases through a system.

reliance (*n*) – the state of depending on something or someone.

replacement (*n*) – something that takes the place of another.

reposition (*v*) – to move something to a new or different location.

reproducibility (*n*) – the ability to be copied or repeated exactly.

robotic arm (*phr*) – a mechanical arm designed for manipulation and precision tasks.

rogue AI (*phr*) – an artificial intelligence system that behaves unpredictably or outside of intended rules, causing harm or malfunction.

rover (*n*) – a mobile robot designed to explore rough terrain.

run like clockwork (*idiom*) – to operate smoothly and predictably.

run out of juice (*idiom*) – to lose power or battery charge.

sensor (*n*) – a part that detects things like light, sound, or walls.

speaker (*n*) – an output device that allows robots to produce sound or speech.

solenoid (*n*) – an electromechanical device that converts electrical energy into linear motion.

steam-propelled (*adj*) – driven or powered by steam.

subset (*n*) – a smaller group that is part of a larger category.

stationary (*adj*) – not moving, fixed in one place.

throw a wrench in the works (*idiom*) – to disrupt or complicate a process unexpectedly.

tracks (*n*) – continuous belts (like on tanks) used for robots needing better stability.

valve (*n*) – a device that controls the flow of liquid or gas in a system.

well-being (*n*) – a state of being healthy, happy, and comfortable.

wheels within wheels (*idiom*) – a complex system with many interconnected parts.

Unit 2 Autonomous Vehicles

5G connectivity (*phr*) – the use of fifth-generation mobile networks that offer fast, real-time data transmission for vehicle communication.

accuracy (*n*) – the degree to which something is correct or exact.

ADAS (*abbr*) – Advanced Driver-Assistance Systems; technologies in vehicles that assist drivers in driving and parking tasks.

AV (*abbr*) – autonomous vehicle; a vehicle capable of driving itself without human input.

avoidance (*n*) – the act of preventing danger or obstacles.

climax (*n*) – the most intense or important point in a process.

complement (*v*) – to add to something in a way that improves or completes it.

comprise (*v*) – to consist of or be made up of different parts or elements.

confuse (*v*) – to make someone uncertain or unable to think clearly.

collision (*n*) – a crash or impact between two or more moving objects or vehicles.

consortium (*n*) – a group of organizations working together on a joint project.

counterpart (*n*) – someone or something that has the same function in a different context.

cruise control (*phr*) – a system that keeps a car moving at a constant speed without the driver pressing the accelerator.

crunch the numbers (*idiom*) – to do complex calculations or data analysis.

curved (*adj*) – not straight; having a smooth, rounded shape.

DARPA (*abbr*) – Defense Advanced Research Projects Agency; a U.S. organization that funds cutting-edge technology, including early autonomous vehicle research.

data theft (*phr*) – the illegal copying or removal of digital information.

designated (*adj*) – officially assigned or marked for a specific purpose.

duplicate (*v*) – to make an exact copy of something.

detect (*v*) – to discover or identify the presence of something using sensors or devices.

determine (*v*) – to find out or decide something through calculation or reasoning.

distinguish fact from fiction (*phr*) – to recognize what is real and separate it from what is imagined or exaggerated.

embedded (*adj*) – fixed firmly within something else.

emergency braking (*phr*) – a safety feature that automatically stops the vehicle to avoid a collision.

encryption (*n*) – the process of converting data into a secure code to prevent unauthorized access.

facilitate (*v*) – to make a process easier or more efficient.

fade into the background (*phr*) – to become less noticeable or less important.

fraud (*n*) – the crime of using deception to gain money or benefits.

FSD (*abbr*) – full self-driving; a level of vehicle automation where the car can drive itself in all conditions without human input.

gain access (*phr*) – to successfully enter or reach data, systems, or places.

green light (*idiom*) – approval to proceed with a plan or project.

groundwork (*n*) – the basic foundation for further development.

hinder (*v*) – to slow down progress.

hit a speed bump (*idiom*) – to meet a minor setback or obstacle.

human intervention (*phr*) – manual control or action taken by a person.

hyperloop (*n*) – a proposed high-speed transportation system where pods travel through low-pressure tubes using magnetic levitation.

intercept (*v*) – to stop or catch something before it reaches its destination.

interfere (*v*) – to get in the way of or disturb a process or action.

intersection (*n*) – a point where two or more roads cross each other.

intervention (*n*) – involvement in a situation to change the result.

just a show (*idiom*) – something designed to impress or entertain but not truly functional or meaningful.

LiDAR (*n*) – a technology that uses laser light to measure distances and create detailed maps.

make sense (*phr*) – to be logical, understandable, or reasonable.

manoeuvre (*v*) – to move or guide something carefully and skillfully, especially in tight spaces.

obstacle (*n*) – an object that blocks the way or needs to be avoided.

odometry (*n*) – the use of data from motion sensors to estimate a vehicle's position over time.

off the beaten track (*idiom*) – to take an innovative approach.

out of gas (*idiom*) – to run out of energy or resources.

outperform (*v*) – to do better than someone or something else in performance or results.

oversee (*v*) – to supervise or manage a task, project, or team.

pedestrian (*n*) – a person walking in an area where vehicles also travel.

personal pod (*phr*) – a small self-driving vehicle designed to transport one or a few passengers.

point cloud (*n*) – a collection of data points in 3D space, often used to represent objects or environments in mapping and navigation.

put the pedal to the metal (*idiom*) – to accelerate effort or speed up development.

rear (*n*) – the back part of a vehicle or object.

road sign (*n*) – a sign placed on or beside a road to give information or instructions to drivers.

road to nowhere (*idiom*) – a project or tech leading to no useful outcome.

self-driving car (*phr*) – a vehicle that can operate without human control.

self-propelled (*adj*) – able to move or operate without external force.

semi-autonomous (*adj*) – having some ability to operate independently but still with some control.

shift gears (*idiom*) – to change approach or strategy suddenly.

SLAM (*abbr*) – Simultaneous Localization and Mapping; a process where a vehicle builds a map of an unknown environment while keeping track of its own location.

smoke and mirrors (*idiom*) – a misleading or deceptive explanation or display meant to hide the truth.

spring (*n*) – a piece of metal that returns to its original shape after being pressed or stretched.

steering (*n*) – the control of a vehicle's direction using a wheel or similar mechanism.

V2X (*abbr*) – Vehicle-to-Everything; a communication system that allows vehicles to interact with other vehicles, infrastructure, and networks.

vulnerable (*adj*) – open to harm, attack, or damage; unprotected.

Unit 3 Mind-control Technologies

alert (*v*) – to cause someone to become attentive or watchful.

Alpha brainwave frequency (*phr*) – a type of brainwave associated with relaxed, calm, and meditative states, typically in the 8–12 Hz range.

amplify (*v*) – to make something stronger or louder.

amplitude (*n*) – the strength or height of a signal wave, such as a brainwave.

amputee (*n*) – a person who has lost one or more limbs.

artificial limb (*phr*) – a man-made arm or leg that replaces a missing one.

attention tracking (*phr*) – monitoring where a person's focus or mental engagement is directed.

biomarker (*n*) – a measurable indicator of a biological condition or process.

blood flow (*n*) – the movement of blood through the body or brain.

brain chip (*phr*) – a small electronic device implanted in the brain to monitor or stimulate neural activity.

brain-computer interface (*phr*) – a system that allows direct communication between the brain and an external device.

brain-monitoring (*adj*) – related to tracking or observing brain activity and signals.

brain-sensing (*adj*) – capable of detecting and monitoring brain signals or activity.

brainwave (*idiom*) – a sudden clever idea or breakthrough.

brainwave-reading cap (*phr*) – a wearable device with sensors that detect brainwave activity.

calibrate (*v*) – to adjust or fine-tune a device for accurate measurement or performance.

circadian rhythm (*phr*) – the natural, internal process that regulates the sleep-wake cycle over a 24-hour period.

cognitive (*adj*) – related to thinking, understanding, learning, and remembering.

cognitive fatigue detection (*phr*) – the process of identifying when a person's mental performance is declining due to tiredness.

cognitive impairment (*phr*) – a condition in which a person has trouble remembering, learning new things, or making decisions.

controller (*n*) – a device used to operate or direct another machine or system.

cortex (*n*) – the outer layer of the brain, responsible for many higher-level brain functions such as thinking and movement.

disabilities (*n*) – physical or mental conditions that limit a person’s movements, senses, or activities.

disrupt (*v*) – to interrupt or disturb the normal function or flow of something.

electroencephalography (EEG) (*n*) – a technique for recording electrical activity of the brain using sensors placed on the scalp.

EEG-driven commands (*phr*) – actions or instructions that are triggered by interpreting brain signals measured through EEG.

EEG headset (*phr*) – a wearable device that uses sensors to monitor brain activity through electroencephalography.

exoskeleton (*n*) – a wearable robotic structure that supports and enhances body movement, used for rehabilitation or assistance.

flow state (*phr*) – a mental state of complete focus and immersion in an activity, often linked with high performance.

grip strength (*phr*) – the force a person can apply when holding or squeezing something with their hand.

haptic feedback (*phr*) – physical sensations, such as vibrations, used to give feedback from a device to the user.

headband (*n*) – a wearable band worn around the head, often used to hold brain sensors in place.

hurdle (*n*) – a difficulty or obstacle that must be overcome.

invasive (*adj*) – involving entry into the body, often through surgery or implants.

intelligible (*adj*) – able to be understood.

jet lag (*n*) – the feeling of tiredness and confusion after traveling across time zones.

lose your marbles (*idiom*) – to act irrationally, often due to tech overload.

magnetic resonance imaging (MRI) (*n*) – a non-invasive imaging technique that uses magnetic fields to create detailed pictures of the brain and other organs.

meditative state (*phr*) – a calm and focused mental condition often achieved through relaxation or mindfulness.

mental effort calibration (*phr*) – the adjustment of systems based on how much mental energy a user is exerting.

metric score (*phr*) – a numerical measurement used to evaluate performance or results.

mind the gap (*idiom*) – to be cautious about a difference or risk.

mindfulness (*n*) – a mental practice of being fully present and aware of the current moment.

mind over matter (*idiom*) – using mental focus to overcome physical or technical challenges.

mobility (*n*) – the ability to move freely and easily.

mood-based (*adj*) – influenced by or responsive to a person’s emotional state.

near-infrared light (*phr*) – a type of light used in some brain-monitoring technologies to measure blood flow and oxygen levels.

neural implant (*phr*) – a device placed inside the brain or nervous system to record or stimulate neural activity.

neural interface hubs (*phr*) – central systems that connect the brain to external devices for communication and control.

neurogadget (*n*) – a wearable or portable device that interacts with or monitors brain activity.

neurons (*n*) – nerve cells that transmit information through electrical and chemical signals.

neuroprosthetics (*n*) – artificial devices that replace or enhance the function of a damaged nervous system or brain region.

neuroscience (*n*) – the scientific study of the brain and nervous system.

neurofeedback (*n*) – a technique that trains individuals to control brain activity by providing real-time feedback.

non-invasive (*adj*) – not involving cutting into the body.

out of your mind (*idiom*) – crazy or unbelievable, often about futuristic tech.

oxygenation (*n*) – the process of adding oxygen to the body or brain tissues.

paralyzed limb (*phr*) – an arm or leg that has lost movement or feeling due to injury or illness.

pick someone’s brain (*idiom*) – to ask for advice or insights, often tech-related.

psychedelic patterns (*phr*) – vivid, often colorful visual effects or brain activity patterns resembling those seen under altered states of consciousness.

rehabilitation (*n*) – the process of recovery and regaining function after injury or illness.

signal interpreter (*phr*) – a device or program that translates brain signals into actions or commands.

sleep cycles (*phr*) – the repeating patterns of light and deep sleep that occur throughout the night.

sleep hygiene (*phr*) – healthy habits and practices that support good quality sleep.

spinal cord (*n*) – the bundle of nerves inside the spine that sends signals between the brain and the body.

spinal injury (*phr*) – damage to the spinal cord that can affect movement, sensation, or bodily functions.

struggle (*v*) – to have difficulty doing something or coping with a situation.

wander (*v*) – to move without a clear purpose or direction; often refers to attention or thoughts.

wheelchair (*n*) – a chair on wheels used by individuals who cannot walk.

white paper (*n*) – an official report that explains a technology or solution.

wired differently (*idiom*) – to think or function in an unconventional way.

wristband (*n*) – a wearable band worn around the wrist.

Unit 4 Digital Citizenship

ALL CAPS (*phr*) – writing in all capital letters, often seen as shouting or overly aggressive in online communication.

audit profiles (*phr*) – to review and evaluate personal online accounts for privacy, security, and content.

boundaries (*n*) – limits that define acceptable behavior or personal space in communication.

caught red-handed (*idiom*) – to be caught in the bad acts.

circulate (*v*) – to spread or pass content around online.

clean slate (*idiom*) – a fresh start, often digitally after a breach or bad behavior.

clickstream (*n*) – a record of the web pages a user visits, showing their browsing path and behavior.

code snippet (*phr*) – a short piece of reusable code shared for reference or discussion.

concise (*adj*) – using few words while being clear and effective.

consent (*n*) – permission for something to happen or be done.

cringeworthy (*adj*) – embarrassing or awkward to the point of discomfort.

curate posts (*phr*) – to carefully select and organize online content before sharing.

cyberbullying (*n*) – using digital platforms to harass, threaten, or harm others repeatedly.

cyberstalking (*n*) – using the internet to repeatedly follow, monitor, or harass someone.

digital footprint (*phr*) – the record of a person’s online activity, including posts, searches, and interactions.

distracting (*adj*) – taking attention away from the main focus or task.

doxxing (*n*) – publishing someone’s personal information online without their consent.

encourage (*v*) – to support or motivate someone to do something.

erase (*v*) – to completely remove digital content or data.

escalate (*v*) – to make a problem or situation become more serious or intense.

exclusion (*n*) – deliberately leaving someone out of an online group or activity.

fish in troubled waters (*idiom*) – to take advantage of chaos.

flaming (*n*) – posting angry or insulting messages in online discussions.

geotagging (*n*) – adding location information to digital content such as photos or posts.

ghosting (*n*) – suddenly cutting off all communication with someone online without explanation.

harassment (*n*) – repeated and unwanted behavior that causes fear, distress, or discomfort online.

hateful (*adj*) – expressing strong dislike or hostility, targeted toward specific individuals.

humiliating (*adj*) – causing someone to feel ashamed or embarrassed, often in public.

impersonation (*n*) – pretending to be someone else online, often to deceive or cause harm.

inappropriate (*adj*) – not suitable or acceptable for a particular context.

indexed (*adj*) – stored and listed by search engines, making online content easier to find.

intentional (*adj*) – done on purpose; planned or deliberate.

judgment (*n*) – the ability to make considered and wise decisions or opinions.

keep your nose clean (*idiom*) – to stay out of trouble.

leak (*v*) – to accidentally or intentionally reveal private or sensitive information.

lurking (*v*) – reading online discussions or content without actively participating.

maintain (*v*) – to keep something at a desired level or state.

mood swings (*n*) – rapid or intense changes in emotional state.

netiquette (*n*) – the set of polite and respectful rules for communicating online.

nudge (*v*) – to gently prompt or remind someone to take an action.

offensive (*adj*) – likely to upset or insult people; disrespectful or inappropriate.

off-topic (*adj*) – not related to the main subject of conversation.

online credibility (*phr*) – the level of trust and reliability a person or source has in digital spaces.

outing (*n*) – revealing someone’s private information or identity online without their permission.

overpost (*v*) – to post too frequently in online discussions, making it hard for others to participate.

oversharing (*n*) – revealing too much personal information online without considering the consequences.

permanent (*adj*) – lasting forever or difficult to remove, especially regarding online content.

profiling (*n*) – collecting and analyzing data about a person’s behavior or preferences to make assumptions or decisions.

pull requests (*phr*) – proposals to merge changes into a codebase.

push someone's buttons (*idiom*) – deliberately annoy or provoke someone

responsibly (*adv*) – in a careful and ethical way, showing consideration for others and consequences.

rumours (*n*) – unverified or false information spread online.

scrub (*v*) – to thoroughly remove unwanted or outdated digital content.

speak up (*phr v*) – to express an opinion or take a stand, especially in defense of others.

status indicators (*phr*) – symbols or labels showing a person’s availability or activity status online (e.g., “away,” “busy”).

stir the pot (*idiom*) – to provoke or cause trouble, often in online forums.

targeted (*adj*) – directed at a specific person or group.

threads (*n*) – sequences of messages or comments grouped by topic in online discussions.

tone awareness (*phr*) – the ability to understand and adjust how one’s message sounds to others.

tone-deaf (*adj*) – lacking awareness of how one’s message may come across.

trail (*n*) – a series of digital traces or records left behind by a user’s online activity.

trolling (*n*) – deliberately posting offensive or disruptive content to provoke reactions.

typing delay (*phr*) – a noticeable pause between messages due to someone typing.

unsecured (*adj*) – not protected by passwords, encryption, or safety measures.
uploading (*v*) – transferring files, images, or information from a device to the internet or a cloud platform.
upstander (*n*) – a person who takes action to support someone who is being mistreated or bullied online.
walk on eggshells (*idiom*) – to act cautiously.

Unit 5 Bioinformatics

AI-based diagnosis (*phr*) – the use of artificial intelligence to identify diseases based on medical data.
algal biofuels (*phr*) – renewable fuels made from algae, considered cleaner than other biofuels.
assemble a jigsaw puzzle (*idiom*) – to put many small pieces together to understand a complex situation or solve a problem.
biodegradable (*adj*) – capable of breaking down naturally in the environment.
bioethics (*n*) – the study of ethical issues in biology and medicine.
biohybrid robot (*phr*) – a robot that combines biological tissues or cells with synthetic materials.
biological sample (*phr*) – material taken from living organisms, such as blood, saliva, or tissue, used for analysis.
biological tissue (*phr*) – groups of cells in the body that perform specific functions, such as muscle or skin.
biomarker (*n*) – a biological indicator used to detect or monitor a disease or condition.
biomanufacturing (*n*) – the use of biological systems to produce materials, drugs, or chemicals.
bioremediation (*n*) – the use of living organisms to clean up environmental pollution.
building blocks (*idiom*) – fundamental components.
cardiac arrest (*phr*) – a sudden stopping of the heart's function.
clinical trial (*n*) – a research study that tests the safety and effectiveness of a medical treatment in people.
communicate instructions (*phr*) – to give commands or input to a computer so it performs specific tasks.
community science project (*phr*) – a scientific study that involves the public in collecting or analyzing data.
compile all data (*phr*) – to collect and organize all available information into a single set.
computational model (*phr*) – a computer program that simulates how something works in real life.
corn-based feedstocks (*phr*) – crops like corn used to produce early-generation biofuels.
crack the code (*idiom*) – to solve a complex problem.
CRISPR (abbr) – Clustered Regularly Interspaced Short Palindromic Repeats; a technology that can edit parts of DNA to correct mutations.
cut to the chase (*idiom*) – to get to the point quickly.
data visualization (*phr*) – the use of charts, graphs, and images to make complex data easier to understand.
disease resistance (*phr*) – the ability of an organism to prevent or fight off illness.
DNA sequence (*phr*) – the order of letters (A, T, C, G) in a DNA that carries genetic information.
drug discovery (*phr*) – the process of finding new medicines through research and testing.
endangered species (*phr*) – animals or plants that are at risk of disappearing forever.
epigenetics (*n*) – the study of changes in gene activity that do not involve changes to the DNA sequence itself.
equity (*n*) – fairness and equal access to opportunities or technologies.
expressed genes (*phr*) – genes that are active and produce proteins or RNA in a cell.
feed the device (*phr*) – to input data into a machine for it to process.

forensic science (*phr*) – the application of scientific methods to solve crimes, often using biological evidence.

fossil fuels (*phr*)– traditional energy sources like coal, oil, and gas that pollute the environment.

gene editing (*phr*) – techniques used to change the DNA of living organisms.

gene mapping (*phr*) – the process of finding the location of genes on a DNA.

genetic disorder (*phr*) – a disease caused by a problem in a person’s DNA.

genetic makeup (*phr*) – all the genes that a person or living thing has.

genetic modification techniques (*phr*) – methods used to change the DNA of organisms for specific purposes.

genome (*n*) – the complete set of genetic material in an organism.

GMO (*abbr*) – genetically modified organism; an organism whose DNA has been changed using biotechnology.

grow like weeds (*idiom*) – to expand rapidly.

gut bacteria (*phr*) – microorganisms living in the human digestive system.

high-throughput (*adj*) – able to do many tests or process lots of data very quickly.

immune system (*phr*) – the body’s defense system against infections and diseases.

infectious disease (*phr*) – an illness caused by bacteria, viruses, or other pathogens that can spread from one person to another.

in your DNA (*idiom*) – an inherent trait or skill.

incompatibility (*n*)– a situation where things cannot work well together.

inherited disease (*phr*) – a medical condition passed from parents through genes.

labeling law (*phr*) – a regulation requiring clear information on product packaging, often about ingredients or modifications.

lab-grown (*adj*) – developed or cultured in a laboratory environment.

life-saving information (*phr*) – critical data that can help protect health or prevent death.

make humans tick (*idiom*) – to understand what drives human behavior.

metagenomic (*adj*) – relating to the study of genetic material recovered directly from environmental samples.

microbial community (*phr*) – a group of tiny living things (like bacteria) that live together in the same place.

microbiome (*n*) – the community of microorganisms (such as bacteria, viruses, and fungi) that live in a particular environment, including the human body.

mRNA vaccine (*phr*) – a type of vaccine that uses messenger RNA to instruct cells to produce a protein that triggers an immune response.

muscular dystrophy (*phr*)– a genetic disease causing muscle weakness and loss.

mutation (*n*) – a change in the DNA sequence that can affect how genes work.

mutation detection (*phr*) – the process of finding changes in DNA that may lead to diseases.

needle in a haystack (*idiom*) – something hard to find.

oil spill (*phr*)– a leak of oil into the environment, especially into the sea.

outbreak's journey and timeline (*phr*) – the path and timing of how a disease spreads over time and location.

overlapping chunks (*phr*) – pieces of data that share common parts and can be connected to form a whole.

pandemics (*n*) – global outbreaks of disease that affect large populations across countries or continents.

pattern recognition (*phr*) – the ability of a system to identify trends or repeated structures in data.

personalized medicine (*phr*) – treatment designed to fit a person’s unique genes and health.

pharming (*n*) – the use of genetically modified organisms to produce pharmaceuticals.

plant the seed (*idiom*) – to start a process or idea.

public data set (*phr*) – a collection of data that is freely available for anyone to use.

public perception (*phr*) – the general opinion or understanding that people have about a topic or issue.

rare disease research (*phr*) – the scientific study of illnesses that affect a very small percentage of the population.

recovery time (*phr*) – the period needed for a person to return to health after illness.

regenerative medicine (*phr*) – a branch of medicine focused on repairing or replacing damaged cells, tissues, or organs.

resistance (*n*) – the ability to fight against something.

RNA molecules (*phr*) – strands of ribonucleic acid involved in protein production and gene regulation.

run into (*phr v*) – to experience a problem or unexpected situation.

scientific undertaking (*phr*) – a large project carried out for scientific purposes.

sequence alignment (*phr*) – the process of arranging DNA, RNA, or protein sequences to identify regions of similarity.

self-heal (*v*) – to recover or repair damage automatically without external help.

side effects (*n*) – unwanted symptoms caused by medical treatment.

social responsibility (*phr*) – the ethical duty to act for the benefit of society at large.

sort through (*phr v*) – to examine a large amount of information to find what is important.

species (*n*) – a group of similar organisms that can reproduce and have offspring.

stem cells (*phr*) – special cells that can develop into different types of body cells.

suppress tumor (*phr*) – to prevent or slow the growth of a cancerous mass in the body.

therapeutic target (*phr*) – a specific molecule or gene that a treatment is designed to affect or correct.

treatment plan (*phr*) – a medical strategy created for managing a patient’s condition.

Unit 6 Blockchain Technology

account value (*phr*) – the amount of cryptocurrency or digital assets held in a blockchain account.

auditor (*n*) – a person who checks records or systems for accuracy and compliance.

authentic (*adj*) – real, genuine, and verifiable.

authenticity (*n*) – the quality of being genuine or real.

batch (*n*) – a group of items or data processed or stored together.

Bitcoin (*n*) – the first and most well-known cryptocurrency, used for peer-to-peer digital transactions.

block validation (*phr*) – the process of checking and approving a block of transactions before adding it to the blockchain.

blockchain (*n*) – a decentralized digital ledger that records transactions across multiple computers.

boon (*n*) – a helpful or beneficial thing.

byproduct (*n*) – something that is produced as a result of another process, sometimes unintentionally.

Cardano (*n*) – a blockchain platform that supports smart contracts with a focus on security and sustainability.

cast a vote (*phr*) – to submit a choice officially in an election or decision-making process.

censorship (*n*) – the act of blocking or suppressing content.

certified official (*phr*) – a person or entity authorized to verify and validate information or documents.

chain of blocks (*phr*) – a system where data blocks are linked in time order.

chain of trust (*phr*) – a series of verified and linked records that ensure authenticity and security.

chain reaction (*idiom*) – a series of events triggered by one action.

cold wallet (*phr*) – a cryptocurrency wallet that is not connected to the internet, offering better security.

collectibles (*n*) – unique items valued for their rarity.

confirm ownership (*phr*) – to verify who legally or digitally owns an asset or token.

consent (*n*) – permission for something to happen required for data use or transactions.

consume (*v*) – to use up resources such as time, energy, or computing power.

cornerstone (*n*) – a basic or essential part of a system or idea.

counterfeit (*v*) – to make a fake copy of something valuable or official.

cryptocurrency (*n*) – a digital form of money that uses cryptography and blockchain technology for secure transactions.

cryptographic (*adj*) – related to methods for securing information using codes and encryption.

decentralized (*adj*) – not controlled by a single authority; spread across multiple participants or locations.

Delegated Proof of Stake (DPoS) (*phr*) – a system where token holders vote to elect delegates who validate transactions and maintain the blockchain.

determine (*v*) – to figure out or decide something through calculation or reasoning.

digital signature (*phr*) – a cryptographic code used to verify the authenticity and integrity of digital messages or documents.

distinguishable (*adj*) – able to be recognized as different or separate.

distributed (*adj*) – shared across multiple locations or computers.

double-spending (*n*) – a type of fraud in digital currency where the same token is spent more than once.

election (*n*) – the process of voting to choose leaders or make decisions.

eligible (*adj*) – meeting the requirements or qualifications to take part in something.

entry (*n*) – a single record or line in a database or ledger.

equation (*n*) – a mathematical expression used to solve problems or validate blockchain processes.

Ethereum (*n*) – a blockchain platform known for supporting smart contracts and decentralized applications.

exploit a flaw (*phr*) – to take advantage of a weakness or bug in a system for personal gain.

exposure (*n*) – the risk of sharing private or sensitive information.

file integrity (*phr*) – the assurance that a digital file has not been altered or corrupted.

forgery (*n*) – the crime of faking documents, assets, or signatures to deceive others.

fraudulent (*adj*) – intended to deceive or cheat.

fungible token (*phr*) – a digital asset that is interchangeable with others of the same type, like cryptocurrencies.

ghost (*v*) – to suddenly cut off all communication.

golden goose (*idiom*) – a valuable asset.

governance (*n*) – the rules and decision-making processes that guide how a blockchain network operates.

hard-code (*v*) – to write fixed values directly into a program's source code, making them difficult to change.

hardware wallet (*phr*) – a physical device that stores cryptocurrency offline for enhanced protection.

hash (*n*) – a fixed-length string of characters generated by a cryptographic algorithm, used to represent data securely.

hold the balance (*phr*) – to have a specific amount of cryptocurrency or tokens in a digital wallet.

hot wallet (*phr*) – a digital wallet connected to the internet, used for quick access to cryptocurrencies.

immutability (*n*) – the quality of being unchangeable or permanent; once data is recorded on a blockchain, it cannot be altered.

in a nutshell (*idiom*) – in brief or in simple terms.

in bulk (*phr*) – in large amounts or quantities.

in the black (*idiom*) – profitable.

independent nodes (*phr*) – separate computers in a blockchain network that operate without centralized control.

insurance claim (*phr*) – a formal request for payment under an insurance policy.

intermediary (*n*) – a person or organization that acts between two others to build trust.

layer (*n*) – a level or component within a blockchain system that handles specific tasks (e.g., Layer 1 for base blockchain, Layer 2 for speed improvements).

ledger (*n*) – a record of transactions or changes kept over time.

lock and key (*idiom*) – a secure system.

meme coin (*n*) – a cryptocurrency that originated from an internet joke or meme, often with little practical use.

mining (*n*) – the process of validating blockchain transactions and adding them to the ledger, often rewarded with cryptocurrency.

Monero (*n*) – a cryptocurrency focused on privacy and untraceable transactions.

network outage (*phr*) – a failure in a computer or internet network.

node (*n*) – a computer in a blockchain network that stores a copy of the ledger and helps verify transactions.

non-destructive (*adj*) – keeping old data and adding new changes without erasing history.

non-fungible token (*phr*) – a unique digital asset that cannot be exchanged one-for-one with another, often used for art or collectibles.

notary (*n*) – a person or service that officially verifies the authenticity of documents; blockchain can serve a similar function.

obsolete (*adj*) – no longer useful or up to date; replaced by newer technology.

opaque (*adj*) – not transparent or easy to understand; lacking visibility or clarity.

paper ballot (*phr*) – a printed voting form used in traditional elections.

paper wallet (*phr*) – a printed record of a cryptocurrency address and private key, used for offline storage.

pass the buck (*idiom*) – to shift responsibility.

peer-to-peer (*adj*) – sharing data directly without a central server.

pending transaction (*phr*) – a transaction that has been submitted but not yet confirmed or added to the blockchain.

permissioned blockchain (*phr*) – a blockchain where only authorized users can participate in validating or viewing transactions.

point of failure (*phr*) – a part of a system where a single error can cause the whole system to stop working.

predefined (*adj*) – set or decided in advance.

probability (*n*) – the likelihood or chance of an event happening.

Proof of Authority (PoA) (*phr*) – a blockchain validation method that relies on approved and trusted validators rather than solving complex puzzles.

Proof of Burn (PoB) (*phr*) – a blockchain protocol where users destroy ("burn") cryptocurrency to show commitment and earn the right to validate blocks.

Proof of Space (*phr*) – a consensus method that uses disk storage space as a resource for mining and validation instead of computing power.

Proof of Stake (PoS) (*phr*) – a consensus mechanism where validators are chosen based on how much cryptocurrency they hold and are willing to "stake" as collateral.

Proof of Work (PoW) (*phr*) – a consensus method where participants solve complex puzzles to validate transactions and earn rewards.

proprietary software (*phr*) – software owned by a company or individual, with restrictions on use and modification.

public access (*phr*) – the ability for anyone to view or use blockchain data or services.

public trust (*phr*) – the confidence people have in the fairness, honesty, or reliability of a system or institution.

publicly viewable (*adj*) – visible and accessible to anyone, without restrictions.

reach consensus (*phr*) – to agree on the validity of a transaction or decision across a network.

real-world pilot (*phr*) – a small-scale trial of a system or technology tested in actual conditions before full implementation.

recordkeeping (*n*) – the act of storing and maintaining accurate records, especially transaction data.

refund (*n*) – a return of money to the payer, especially when a product or service is canceled or defective.

registry updates (*phr*) – changes made to a digital record or database, often involving ownership or status.

replicate (*v*) – to copy or reproduce something exactly.

ring signature (*phr*) – a type of digital signature that hides the sender’s identity by mixing it with other possible signers.

Ripple (*n*) – a digital payment protocol and cryptocurrency designed for fast, low-cost international transactions.

self-adjusting (*adj*) – able to automatically change or adapt based on conditions without human input.

self-executing (*adj*) – automatically carrying out actions when certain conditions are met.

small-scale (*adj*) – limited in size, number, or scope.

smart contract (*n*) – a self-executing digital agreement stored on a blockchain that automatically carries out terms when conditions are met.

smoke and mirrors (*idiom*) – deceptive tactics.

spot (*v*) – to notice or identify something, such as errors or unusual activity.

stake (*n*) – a value or amount of cryptocurrency held or risked in a blockchain system, often used in validation processes.

stealth address (*phr*) – a one-time address used in privacy-focused blockchain systems to hide the recipient’s identity.

sync across the network (*phr*) – to update and match data on all nodes in a blockchain network to ensure consistency.

tamper (*phr*) – to illegally or dishonestly change data or outcomes.

tamper-proof (*adj*) – designed to prevent unauthorized changes or alterations.

tampering (*n*) – unauthorized changes or interference, especially with data or systems.

tighten the screws (*idiom*) – to increase security or control.

timestamp (*n*) – a digital record showing the exact date and time a transaction occurred.

token (*n*) – a digital unit of value or utility used in a blockchain ecosystem.

transaction fee (*phr*) – a small payment made to process or validate a transaction on a blockchain.

transfer date (*phr*) – the specific date when ownership or control of a digital asset changes hands.

transferable (*adj*) – able to be passed from one person or entity to another.

transparent (*adj*) – open and easy to see or verify; not hidden.

underlying (*adj*) – forming the base or foundation of something, such as technology behind a system.

validator node (*phr*) – a computer in a blockchain network that checks and approves transactions and blocks.

vending machine (*n*) – a machine that sells items automatically.

verify (*v*) – to check or confirm that something is true or correct.

voter privacy (*phr*) – the protection of a person’s identity and choices in an election system.

virtual assets (*phr*) – digital items that hold value, such as cryptocurrencies or tokens.

Unit 7 Sustainable Computing

AI-powered (*adj*) – enhanced or controlled by artificial intelligence technology.

air circulation (*phr*) – the movement of air to help cool devices and prevent overheating.

awareness (*n*) – knowledge or understanding of a particular issue or situation.

biodegradable (*adj*) – capable of breaking down naturally without harming the environment.

burn the candle at both ends (*idiom*) – to overwork.

carbon emissions (*phr*) – the release of carbon dioxide into the atmosphere, often from energy use or manufacturing.

circuit board (*phr*) – a flat piece of material that holds electronic components and connects them through circuits.

circular economy (*phr*) – an economic model focused on reusing, recycling, and repairing to minimize waste.

closed-loop system (*phr*) – a production system where waste is collected and reused in the same cycle.

cloud computing (*phr*) – storing and accessing data and programs over the internet instead of on local devices.

collection point (*phr*) – a designated location where used items or waste can be dropped off for recycling or disposal.

comprise (*v*) – to consist of or be made up of specific parts or elements.

copper (*n*) – a valuable metal commonly used in electronics and wiring.

CPU throttling (*phr*) – reducing the processor's speed to lower heat and power usage.

cradle-to-cradle (*phr*) – a sustainable design approach where products are fully reusable or recyclable, creating no waste.

de-manufacturing (*n*) – carefully taking apart products to recover reusable or recyclable parts.

deep sleep (*phr*) – a state where a computer uses minimal power and takes longer to wake up than from regular sleep.

disassembly (*n*) – the process of taking apart a product, often to recycle or repair its components.

discarded electronics (*phr*) – electronic devices that have been thrown away or are no longer in use.

digital product passport (*phr*) – a digital record that tracks the origin, use, materials, and recyclability of a product.

discard (*v*) – to throw something away because it is no longer useful or wanted.

display off (*phr*) – a setting that turns off the screen to save energy while keeping the system running.

dispose (*v*) – to get rid of something, especially waste or unwanted items.

downcycle (*v*) – to recycle materials into products of lower quality or value.

drop in the bucket (*idiom*) – a small effort compared to the problem.

efficiency ratio (*phr*) – the amount of computing power compared to energy used.

energy audit (*phr*) – an analysis to find where and how to save energy in IT systems.

energy-efficient (*adj*) – using less energy to perform the same task, reducing environmental impact.

energy recovery (*phr*) – reusing waste heat or power generated during computing.

energy saver (macOS) (*phr*) – a macOS feature that helps reduce energy usage through settings like display sleep.

Energy Star-certified (*phr*) – labeled as energy-efficient according to standards set by the Energy Star program.

equate (*v*) – to consider one thing the same as or similar to another.

extend (*v*) – to make something last longer or continue further.

extract (*v*) – to remove something, often natural resources, from the earth.

e-waste (*n*) – discarded electronic devices or components, such as phones or computers.

facility (*n*) – a place or building used for a specific purpose, such as manufacturing, storage, or recycling.

fall short (*phr v*) – to fail to meet expectations or a required standard.

green computing (*phr*) – environmentally responsible use of computers and related technology.

hazardous (*adj*) – dangerous or harmful, especially to people or the environment.

heat waste (*phr*) – unwanted heat produced by electronic devices that must be managed or reduced.

hibernate mode (*phr*) – a power-saving state where the system saves its memory to disk and shuts down almost completely.

idle power draw (*phr*) – the energy a device uses when it's on but not doing tasks.

incinerator (*n*) – a facility used to burn waste materials at high temperatures.

landfill (*n*) – a site where waste is buried in the ground.

lead exposure (*phr*) – contact with lead, a toxic metal that can cause serious health problems.

leave no stone unturned (*idiom*) – to explore every option.

linear economy (*phr*) – a traditional model based on using resources once and discarding them.

liquid cooling (*phr*) – a method of reducing heat in computers by using liquid instead of air.

live on the margins (*phr*) – to survive with limited resources or opportunities.

low-power mode (*phr*) – a setting that reduces energy use by turning off non-essential functions.

make a dent (*idiom*) – to have a noticeable but limited impact on a big issue.

modular design (*phr*) – building products with replaceable or upgradeable parts.

natural airflow (*phr*) – the movement of air without mechanical assistance, used for passive cooling.

obsolescence (*n*) – the state of being outdated or no longer useful, often due to new technology.

on-demand computing (*phr*) – running systems only when needed to avoid wasting power.

open-loop recycling (*phr*) – waste from one system is recycled and used in a different one.

ore (*n*) – natural rock or sediment that contains valuable minerals, such as metals.

pluck out (*phr v*) – to remove something quickly or forcefully, often by hand.

power cap (*phr*) – a limit set to reduce how much energy a server or device can use.

power consumption (*phr*) – the amount of electricity used by a device or system.

power saver mode (Windows) (*phr*) – a Windows feature that minimizes power use by lowering performance and screen brightness.

product-as-a-service (*n*) – a business model where customers pay to use a product instead of owning it.

proof of concept (*phr*) – a demonstration that shows an idea or design works in practice.

ramification (*n*) – a possible consequence or result of an action.

recyclable (*adj*) – capable of being processed and used again as raw material.

refurbish (*v*) – to repair and update used equipment so it can be used again.

removable (*adj*) – able to be taken out or separated from something else, such as a battery or part.

repairability (*n*) – the ease with which a product can be fixed or restored to working condition.

retain (*v*) – to keep or hold on to something.

reusable (*adj*) – able to be used again, often after cleaning or repairing.

run on fumes (*idiom*) – to operate with minimal resources, often inefficiently.

safety equipment (*phr*) – tools or clothing used to protect a person from harm while working, especially with hazardous materials.

scarcity (*n*) – a lack or shortage of something, especially resources.

server farm (*phr*) – a large group of servers often used by data centers.

set a fire (*phr*) – to cause something to start burning, either accidentally or intentionally.

sleep mode (*phr*) – a low-power state that keeps a computer ready for quick use without fully turning it off.

smart labeling (*phr*) – technology (like QR codes or chips) added to products to provide information about reuse, recycling, or repair.

solar-powered (*adj*) – running on energy from sunlight instead of conventional electricity.

sow the seeds (*idiom*) – to initiate.

substantial (*adj*) – large in amount, size, or importance.

survey (*n*) – a set of questions used to gather information or opinions from people.

susceptible (*adj*) – easily affected or harmed by something.

sustainable phone (*phr*) – a mobile device designed with environmentally friendly materials and features like easy repair or recycling.

take-back program (*phr*) – a system where manufacturers accept old products for recycling or safe disposal.

thermal management (*phr*) – controlling heat in computers to avoid damage or power waste.

turn over a new leaf (*idiom*) – to make a fresh, positive change.

under warranty (*phr*) – still covered by a guarantee for repair or replacement by the manufacturer.

urban mining (*phr*) – the process of recovering valuable materials from electronic waste found in cities.

upcycle (*v*) – to transform waste materials into new products of higher value or quality.

upgradable (*adj*) – able to be improved or updated with new components or software.

virtualization (*n*) – the use of software to run multiple systems or applications on a single physical machine.

zero-emission data center (*phr*) – a facility that runs fully on renewable or clean energy sources.

zero-waste (*n*) – a system that aims to eliminate all waste through reuse and efficient design.

Unit 8 The Future of Computing

adaptive (*adj*) – able to change or adjust in response to different conditions or users.

ahead of its time (*idiom*) – tech too advanced for current use.

AI co-pilot (*phr*) – an AI assistant that supports users in tasks like coding, writing, or problem-solving.

angle (*n*) – the direction or orientation from which something is viewed or projected.

assert (*v*) – to state or declare something confidently.

augmented intelligence (*phr*) – a collaboration between humans and AI that enhances human capabilities rather than replacing them.

badge (*n*) – a digital reward or marker of achievement used in gamified systems.

bend (*v*) – to change direction when describing flexible screens or light paths.

bi-directional (*adj*) – functioning in two directions, such as communication or data flow.

by leaps and bounds (*idiom*) – very quickly and dramatically.

code completion (*phr*) – a feature in programming tools that predicts and fills in code as the user types.

compiler (*n*) – a program that translates code written by humans into instructions a computer can understand.

context-aware (*adj*) – able to recognize and respond to the surrounding situation.

cryogenics (*n*) – the science of producing very low temperatures, essential for stabilizing quantum systems.

decoherence (*n*) – the loss of quantum behavior due to interference from the environment, making the system act classically.

depth-sensing (*adj*) – detecting how far objects are from a sensor, often used in 3D cameras.

dilution (*n*) – the process of reducing concentration, often referring to ultra-cold temperatures used in quantum systems.

dimension (*n*) – a measurable extent of space, such as height, width, or depth.

downtime (*n*) – a period when a system or device is not working or is unavailable.

emitter (*n*) – a device that sends out light, sound, or other signals.

emission (*n*) – the release or sending out of light, heat, or other forms of energy.

entire (*adj*) – whole or complete; including everything.

entangle (*v*) – to link qubits so that the state of one instantly affects the other, even at a distance.

factoring (*n*) – breaking a number into its prime components.

field of view (*phr*) – the extent of the observable area that can be seen through a device.

float (*v*) – to appear to stay in the air without visible support.

flat (*adj*) – having a level, two-dimensional surface.

fraction (*n*) – a small part or portion of a whole.

gamification (*n*) – the use of game-like elements in non-game contexts to increase engagement.

gesture recognition (*phr*) – interpreting human movements (e.g. hands) as input commands.

heads or tails (*phr*) – a reference to a coin flip, often used to explain binary or probabilistic outcomes.

herald (*v*) – to signal or announce the beginning of something important.

holographic projection lens (*phr*) – a lens that helps form and direct holographic images into space.

hologram (*n*) – a 3D image created by light beams, often appearing to float in space.

human-in-the-loop (*phr*) – a system design that involves human oversight in automated decision-making.

in mid-air (*phr*) – appearing or existing in open space, not touching any surface.

interference (*n*) – unwanted influence from external factors.

jump the gun (*idiom*) – to act prematurely on tech predictions.

laser emitter (*phr*) – a device that produces a focused beam of light used in scanning or imaging systems.

leaderboard (*n*) – a ranked list showing the top performers in a system or activity.

light at the end of the tunnel (*idiom*) – hope for a breakthrough.

light beam (*phr*) – a narrow stream of light energy.

light field generator (*phr*) – a system that creates a full 3D light pattern for producing lifelike images in space.

logic gate (*phr*) – a basic building block in computing that processes binary input to produce a result.

motion sensor (*phr*) – a device that detects movement and can trigger responses in a system.

natural language processing (*phr*) – a field of AI focused on enabling computers to understand and interact using human language.

noise (*n*) – random disturbances that interfere with the operation of a quantum computer.

notification (*n*) – an alert or message that provides updates or prompts action.

optical waveguide (*phr*) – a structure that directs light through a device, often used in displays and communication.

peer comparison (*phr*) – evaluating performance or progress against that of others.

precision (*n*) – the quality of being exact and accurate.

preventive (*adj*) – intended to stop problems before they occur.

quantum computing (*n*) – a type of computing that uses the principles of quantum physics to process information far faster than traditional computers.

quantum gate (*phr*) – an operation that manipulates qubits, similar to how classical logic gates process bits.

quantum supremacy (*phr*) – the point at which a quantum computer can solve problems no classical computer can solve in a reasonable time.

qubit (*n*) – the basic unit of quantum information, which can represent 0, 1, or both at the same time.

raise the bar (*idiom*) – to set a higher standard.

reflection (*n*) – the bouncing back of light or sound from a surface.

rendering engine (*phr*) – software that generates images or 3D models in real time.

rotate (*v*) – to turn around an axis or center.

sensitive (*adj*) – easily affected by external conditions, such as temperature or vibration.

shielding (*n*) – protection from external influences like heat, light, or electromagnetic radiation.

sky's the limit (*idiom*) – unlimited potential.

spaced repetition (*phr*) – a learning technique that reviews information at increasing intervals to improve memory retention.

spatial (*adj*) – relating to space or the position, area, and size of things within it.

spatial light modulator (*phr*) – a device that controls the light intensity and phase in a display or holographic system.

spotlight (*n*) – a focused beam of light or attention on a particular area, object, or topic.

stability (*n*) – the ability of a system to remain unchanged or unaffected by external influences.

sub-pixel (*n*) – a tiny part of a pixel used to increase image resolution or color accuracy.

superposition (*n*) – a quantum state where a particle can exist in multiple states at once.

surpass (*v*) – to exceed or go beyond something in quality or performance.

telepresence (*n*) – the use of technology to feel or appear present in a distant location.

through the roof (*idiom*) – to increase dramatically.

throughput (*n*) – the amount of data or work processed by a system in a given time.

tip of the iceberg (*idiom*) – just the beginning of a vast potential.

transmission (*n*) – the passing of light, sound, or data from one point to another.

unattainable (*adj*) – impossible to reach or achieve.

viewable (*adj*) – able to be seen or displayed.

visual dashboard (*phr*) – a screen or interface that displays data in a clear, visual format for easy understanding.

waveguide (*n*) – a structure that directs the flow of light or other waves.